

WASP-12b

R-1057

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161220 · created 2026-07-30T11:12:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457742.85431 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.062 +/- 0.027
Transit depth (Rp/Rs)^2	0.39 +/- 0.34 [%]
Orbital Inclination (inc)	83.42 +/- 2.71
Airmass coefficient 1 (a1)	1.112 +/- 0.026
Airmass coefficient 2 (a2)	-0.082 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.37 %
Best Comparison Star	#1 - [81, 68]
Optimal Method	PSF photometry
Transit Duration (day)	0.1165 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.062 ± 0.027 vs 0.1178 (deviation 0.95σ)
Duration fit vs published	0.1165 d vs 0.125 d (deviation 0.58σ)
Mid-time O–C	0.5 min
Depth SNR	1.1
Residual scatter	2.37 %

Light curve

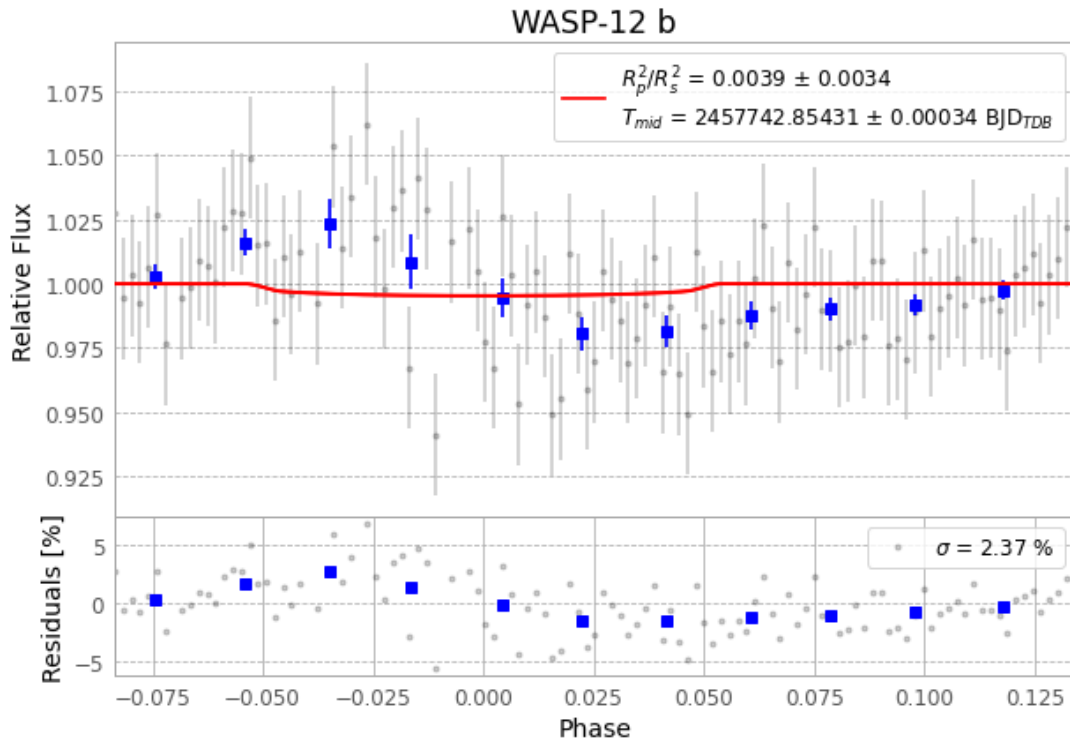


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-19.png (SHA-256 7d57be292e46319e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d95718bfb0c6db4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

115 FITS frames (76 MB), WASP-12161220060912.FITS → WASP-12161220115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161220.log (ba31674b332c...), FinalLightCurve_WASP-12 b_2016-12-19.png (7d57be292e46...), FinalLightCurve_WASP-12 b_2016-12-19.csv (d2ad897bf214...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 11:12Z.